MULTICENTER OBSERVATIONAL COHORT · STROBE · SYNTHETIC DEMO DATA

Early SGLT2 Inhibitor Initiation Within 72 Hours of Acute Heart Failure Admission and 24-Month Cardiorenal Outcomes

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01 Clinical Background & Objective

Acute decompensated heart failure (ADHF) hospitalization marks high vulnerability for recurrent congestion and progressive kidney decline [1]. Whether initiating SGLT2 inhibitors within 72 hours of inpatient stabilization improves 24-month cardiorenal survival across ejection fraction strata remains uncertain.

• **Primary Objective:** Compare 24-month composite cardiovascular death, HF rehospitalization, or ≥40% sustained eGFR decline between early inpatient SGLT2i (≤72 h) vs. standard post-discharge care.

02 Cohort Derivation & Propensity Weighting

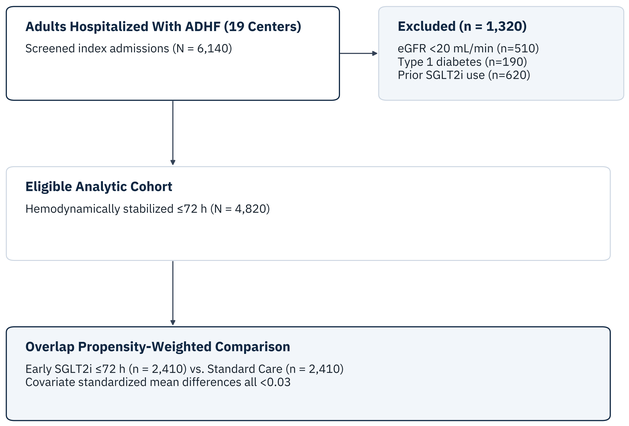


Figure 1. STROBE cohort selection across 19 hospitals (N = 4,820; synthetic cohort).

03 Statistical Architecture

• **Exposure:** Empagliflozin or dapagliflozin started ≤72 h of IV loop diuretic stabilization.

• **Models:** Overlap propensity-weighted Cox models & Fine-Gray competing-risk regression adjusted for age, LVEF, NT-proBNP, baseline eGFR, and background GDMT [2].

Primary 24-Month Cardiorenal Composite

aHR 0.68 (95% CI 0.56–0.82; p = 0.0001)

Early SGLT2i initiation reduced cumulative 24-month event incidence from 29.8% (718/2,410) to 21.4% (516/2,410), corresponding to an absolute risk reduction of 8.4% (NNT = 12).

04 24-Month Event-Free Survival

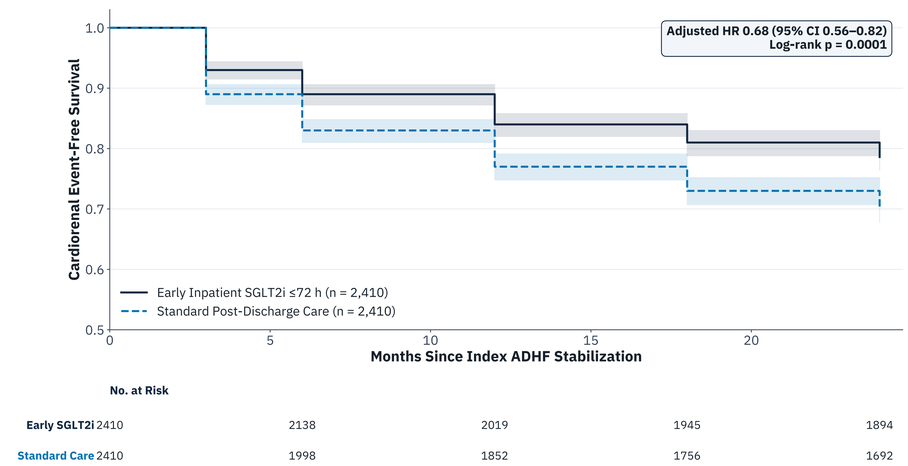


Figure 2. Overlap-weighted Kaplan-Meier cardiorenal event-free survival with shaded 95% CIs and aligned number at risk (N = 4,820).

05 Component & Safety Endpoints

Outcome (24 Months)	Early SGLT2i	Standard	aHR (95% CI)	p-value
Primary composite	21.4%	29.8%	0.68 (0.56–0.82)	0.0001
HF rehospitalization	14.8%	21.9%	0.65 (0.53–0.80)	0.0002
CV mortality	6.9%	8.8%	0.78 (0.61–0.99)	0.041
≥40% eGFR decline	5.2%	7.3%	0.71 (0.55–0.92)	0.009
HFrEF (<40%) stratum	22.1%	31.0%	0.67 (0.54–0.83)	0.0003
HFpEF (≥50%) stratum	20.3%	28.1%	0.70 (0.55–0.89)	0.004
Euglycemic DKA (safety)	0.4%	0.3%	1.31 (0.52–3.30)	0.56

Subgroup p-interaction by LVEF phenotype = 0.64. Euglycemic DKA was exploratory safety outcome.

06 Strengths & Limitations

• **Strengths:** Active-comparator new-user design across 19 centers with adjudicated renal laboratory trajectories.

• **Limitations:** Residual confounding by unmeasured inpatient diuretic responsiveness cannot be excluded; post-12-month adherence relied on dispensation claims.

Clinical Conclusions

Initiating SGLT2 therapy within 72 hours of inpatient ADHF stabilization was associated with a 32% relative hazard reduction in 24-month cardiorenal events across HFrEF and HFpEF phenotypes, without excess ketoacidosis or acute renal injury.

References: [1] Voors AA et al. Nat Med 2022;28:568–574. [2] Li F, Morgan KL, Zaslavsky AM. JASA 2018;113:390–400.

Funding: Synthetic Translational Grant SYN-2026-014 | Correspondence: e.rostova@synthetic-ahn.org | SYNTHETIC DATASET DISCLOSURE: Simulated cohort created strictly for poster skill testing.